Proposal Evaluation Form

EUROPEAN COMMISSION

European Defence Fund (EDF)

Evaluation Summary Report

Call: EDF-2024-LS-RA-CHALLENGE

Type of action: EDF-LS

Proposal number: 101224078

Proposal acronym: SWARMER

Duration (months): 48

Proposal title: Swarming Autonomy in non-peRMissive EnviRonments

Activity: EDF-2024-LS-RA-CHALLENGE

N.	Proposer name	Country	Total eligible costs	%	Grant Requested	%
1	IKNOWHOW SA	EL	879,548.25	17.60%	879,548.25	17.60%
2	DEUTSCHES ZENTRUM FUR LUFT - UND RAUMFAHRT EV	DE	584,985.5	11.71%	584,985.5	11.71%
3	APPLIED INTELLIGENCE ANALYTICS LIMITED	IE	430,081.4	8.61%	430,081.4	8.61%
4	ETHNIKO KENTRO EREVNAS KAI TECHNOLOGIKIS ANAPTYXIS	EL	450,625	9.02%	450,625	9.02%
5	ETRA INVESTIGACION Y DESARROLLO SA	ES	617,423.26	12.36%	617,423.26	12.36%
6	SIGNALGENERIX LIMITED	CY	504,696.01	10.10%	504,696.01	10.10%
7	STOWARZYSZENIE POLSKA PLATFORMA BEZPIECZENSTWA WEWNETRZNEGO	PL	419,709.3	8.40%	419,709.3	8.40%
8	ICARUS SWARMS	FR	480,000	9.61%	480,000	9.61%
9	THALES	FR	629,937.76	12.61%	629,937.76	12.61%
	Total:		4,997,006.48		4,997,006.48	

Abstract:

The evolution of unmanned aerial systems (UAS) is vital for future defence operations, especially as conflict zones increasingly disrupt traditional navigation technologies like GNSS. Adversaries now use advanced jamming and spoofing techniques, leaving forces vulnerable if dependent on GNSS for navigation and coordination. UAS must evolve to operate autonomously in denied environments, executing complex tasks like reconnaissance and coordinated attacks. The SWARMER project addresses these needs by developing a drone swarm capable of independent navigation in GNSS-denied settings, integrating multi-sensor systems and intelligent fusion techniques. This solution ensures drones can adapt to dynamic conditions and cooperate in swarming formations, enhancing defence capabilities, aligning with the EU's 2023 Capability Development Priorities for Ground Combat, Military Mobility, Infrastructure Protection, and Airborne Command. The Ministries of Defence of Greece, Poland, Cyprus and Ireland support the SWARMER proposal, as its expected outcomes align with their operational interests and will benefit their goals and objectives through the acquisition of valuable knowledge.

Evaluation Summary Report

Evaluation Result

Total score: 38.00 (Threshold: 30)

Result of DEFIS evaluation

Score: **38.00** (Threshold: 30 / 45.00 , Weight: -)

Criterion 1 - Excellence (/5,0 - increment of 0,5) – Weight x2
Contribution to excellence or potential of disruption in the defence domain, in particular by showing that the expected results of the proposed action present significant advantages over existing defence products or technologies
Score (/5,0 - increment of 0,5) 3,5

Comments
The proposed concept is highly ambitious, addressing the problem in a holistic manner by covering most of the key aspects of autonomy and swarming research. Details about the main ideas, technologies and methodology are presented in a very comprehensive way. However, while some technologies are described in depth such as swarming, the proposal provides less details on navigation techniques and in demonstrating convincingly how they should be applied in the sensor fusion.

The proposal addresses well the specific challenge, covering target activities and most of the functional requirements as outlined in the call's documentation. It highly matches the expected impact. However, even if the proposal states compliance with GNSS denied navigation, it does not sufficiently clarify the reliance in robust GNSS receivers as the primary source of navigation information in single drone navigation.

The proposal effectively outlines the strategic, technological, and operational advantages it offers. Its potential for disruption lies in the development of an efficient and miniaturized edge AI board, which would ensure accuracy when deploying AI on the target drone. However, it offers limited discussion on how it differs from current systems and their technological capabilities.

Criterion 2 - Innovation (/5,0 - increment of 0,5) – Weight x2
Contribution to the innovation and technological development of the European defence industry, in particular by showing that the proposed action includes ground-breaking or novel concepts and approaches, new promising future technological improvements or the application of technologies or concepts previously not applied in defence sector, while avoiding unnecessary duplication
Score (/5,0 - increment of 0,5) 4,0

Comments
The proposal provides a very good analysis of the literature on the specific elements of innovation. It addresses new promising technological improvements in local communication systems (mesh networks), Lidar-based obstacle detection and avoiding modules, secure communication between unmanned aerial vehicle (UAV) and tactical cloud, and UAV swarm coordination and decision-making areas and in on board computing considering the edge-AI. However, the proposal has moderate innovation potential in the area of autonomous navigation.
The proposal presents a very detailed table with previous or ongoing R&D activities and explains very well how the knowledge assimilated in these projects can be integrated into the solution.
However, for one specific partner AIA, the proposal insufficiently details how it avoids duplication with ongoing projects.

The proposal presents very well a series of technologies that could spin-off to other defence applications, that are reliable and logically explained. Also, the proposal very well explains how each member of the consortium could spin-off to other defence technologies, further developing the field in which they operate.

Criterion 3 - Competitiveness and Growth (/5,0 - increment of 0,5) – Weight x1
Contribution to the competitiveness and growth of defence undertakings throughout the Union, in particular by creating new market opportunities.
Score (/5,0 - increment of 0,5) 4,5

Comments
The proposal presents very well the competitive advantage by enhancing autonomy and resilience in hostile conditions for drone missions and swarming, supported by a scalable and interoperable architecture. It clarifies very well the areas where its solution outperforms existing systems, both in the market and under development. The competitive edge is also very well-justified, particularly in terms of the technologies employed. However, the proposal does not sufficiently demonstrate the competitive advantage in terms of supporting quantitative data or detailed comparison with existing products.

The proposal excellently outlines its potential for accelerating company growth that is considered very high, supported by very good and convincing elements, such as a thorough analysis of both the European and global military UAV markets, highlighting their growth potential and an insightful SWOT analysis. In addition, it defines a clear business model, relying on a market-oriented approach with structured activities and milestones to be developed throughout the project.

The proposal presents a highly professional and sound IPR strategy. The related funding principles, which are already agreed upon by the partners, are highly convincing, promoting collaboration while safeguarding the rights of each partner.

Criterion 4 – Contribution to autonomy and security and defence interest (/5,0 - increment of 0,5) – Weight x1
Contribution to the autonomy of the EDTIB, including by increasing the non-dependency on non-Union sources and strengthening security of supply, and to the security and defence interests of the Union in line with the priorities referred to in Article 3 of EDF Regulation.

Score (/5,0 - increment of 0,5) 4.0

Comments
The proposed action has the potential to contribute to the autonomy of the EDTIB by increasing the Union's industrial and technological non-dependency from third countries. However, insufficient details are given on the current industrial and technological dependence from third countries.

The security of supply is very well addressed and it is clear that the consortium has the relevant actors for the actual implementation of a potential supply chain.

In terms of contribution to CDP agreed by Member States, the consortium identifies clearly some of the defence capability such as: Intelligence, Surveillance, and Reconnaissance (ISR), Operational Autonomy and Resilience in Contested Environments and Cross-Domain Integration and Enhanced Interoperability.

Criterion 5 - Creation of cross-border cooperation (/5,0 - increment of 0,5) – Weight x2
Contribution to the creation of new cross-border cooperation between legal entities established in Member States or associated countries, in particular SMEs and mid-caps with a substantial participation in the action, as recipients, subcontractors or as other legal entities in the supply chain, and which are established in Member States or associated countries other than those where the legal entities cooperating within a consortium which are not SMEs or mid-caps are established.

Score (/5,0 - increment of 0,5) 5,0

Comments
The proposal is excellent in supporting the creation of new cross-border cooperation between legal entities established in Member States and the extension of previously initiated cooperation between some partners.

The proposal provides an excellent analysis of the potential future cooperation opportunities for promoting considering different perspectives such as transfer and market expansion, engagement of SMEs and mid-caps in niche technologies, knowledge exchange and cross-border training initiatives. All these considerations enable significantly the collaboration, the integration of capabilities and the technology sharing for the future.

The proposal excellently demonstrates the added value brought by most of the SMEs involved, including cross-border SMEs. They provide specific expertise in key areas, such as: cloud-enabled processing solutions (IKH); AI-driven analytics and data processing algorithms (AIA); edge computing integration (SG) and swarm coordination algorithms and decentralized decision-making protocols (ICARUS).

Criterion 6 – Quality and efficiency of implementation (/5,0 - increment of 0,5) – Weight x1
The quality and efficiency of the carrying out the action

Score (/5,0 - increment of 0,5) 4,5

Comments
The proposal presents a highly plausible and very well-structured work organization. The Work Breakdown Structure (WBS) is carefully organized, fully aligned with the expected outcomes. All work packages (WPs) within the same phase are designed to run in parallel, ensuring minimal coupling between activities. This approach leads to several advantages: an appropriate timeline for each WP, simplified management, a realistic schedule for achieving the stated objectives, and well-defined interrelations between the work packages and tasks.

The proposal defines proper milestones and deliverables that are closely aligned with the respective work packages and tasks. The milestones are very well balanced in number and carefully designed to capture key information from each project phase. They are measurable, with clear and realistic criteria for achievement. The proposed timeframes for reaching them are both highly convincing and very well-structured. The deliverables are very much suited to describe the expected outcomes.

The proposal sets an appropriate structured management structure, processes and procedures of decision making that are aligned to the project needs and complexity. It refers to applicable standards to be used which are convincing about the quality of the management action. Furthermore, it outlines critical risks for implementation properly. Risk likelihood and impact are realistically assessed. The proposed mitigation measures are very appropriate. However, the proposal does not sufficiently detail the technological risks. For example, AI management in terms of training needs or simulation environment.

The work is properly allocated to most of the partners on the basis of their background. In the action consortium members contributions are mostly complementary. WP leadership and task leadership load are distributed across the consortium members. Work share is well distributed. However, the proposal does not sufficiently describe the role of partner PPHS. In addition, except for the project coordinator - IKH - also the PPHS plays the pure coordinating role. It is not clear the reasons for the allocation to PPHS of managerial task.

Coherence and quality of the financial proposal
Does the financial proposal ensure consistency of the effort for each work package and for each activity per applicant (number of person/months required) and on the technical necessity of the costs concerned for the implementation of the project?
Personnel costs
Is the team proposed by the applicants in the annexes adequate (size and profile) for the implementation of the action?
YES

Are the human resources (staff categories and number of days used in the action) proposed by the applicants adequate to carry out the activities and reach the objectives described in the proposal?
YES

Are the staff costs reasonable? For lump sum grants, are the annual declared staff costs declared below EUR 200 000?
YES

Do the effort / cost of management and coordination of the project exceed 10% of the total effort / eligible costs?
YES

If YES, please provide a short explanation
In terms of the effort project management weights 14% as declared in the part B in the related section to workshare and needs further justification. In this project, the management budget is big enough to cover all managerial activities by the IKH's budget.

Subcontracting
Are subcontractors included in the budget table clearly identified in Part B? Are the "Justification" and "Best value for money" fields complete and clear?
Not applicable

Are there subcontractors for Management and Coordination of the project (typically WP1)?
NO

Are the subcontracting costs included in the budget reasonable and justified for the implementation of the action?
Not applicable

If the percentage of subcontracting costs is higher than 30% of the total applicant's eligible costs, is the provided justification acceptable?
Not applicable

Other costs
Are the number and purpose of trips adequate for the implementation of the action?
YES

Are the number of trips proportional to the duration of the action (more than 1 trip per month per applicant, 2 per month for coordinator)?
YES

Are there more than 2 persons travelling per trip?
YES

If YES, please provide a short explanation
IKW, CERTH and THALES have a few WPs where they proposed 3 people/trip. This is justified by the nature of the project. IKH - project coordinator and Thales for Kick-off Meeting and final project meeting plan to have three-person delegations. This is not justified.

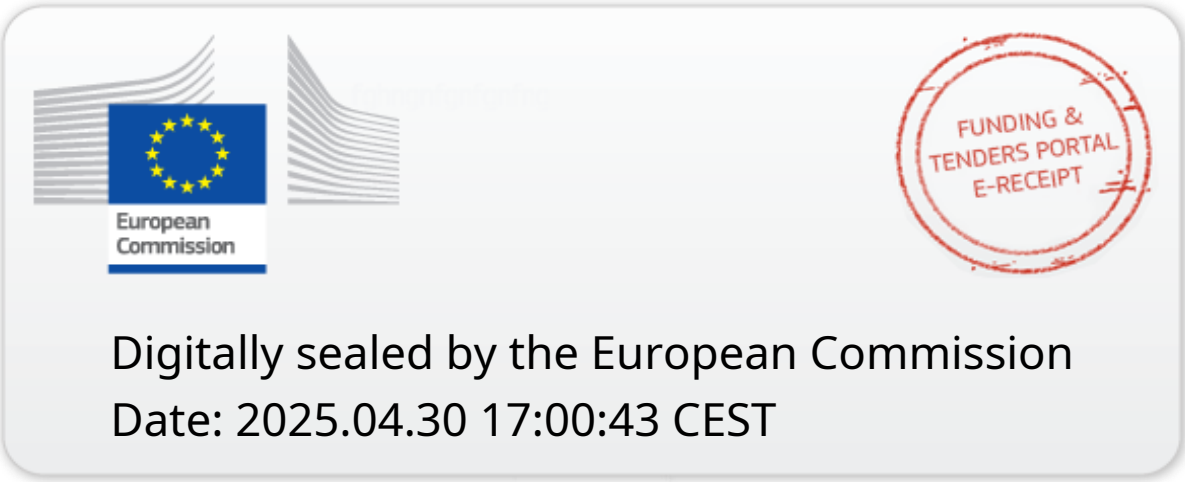
Are the Purchase costs (equipment and other goods, works & services) and the costs included under "Other cost categories" adequate/necessary for the implementation of the action and their cost reasonable?

NO

If NO, please provide a short explanation
The partner ETHRA is asking 60000 euro for other goods. This needs further justification.

If equipment is declared as full costs (only for prototyping activities) are the justification and best value for money fields complete and acceptable?
YES

The Ethics Summary Report will be sent at a later stage.



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